

Microsoft Excel Accessibility for AccessU 2025

How to Create Excel Files that Can Be Read by All People

About Me

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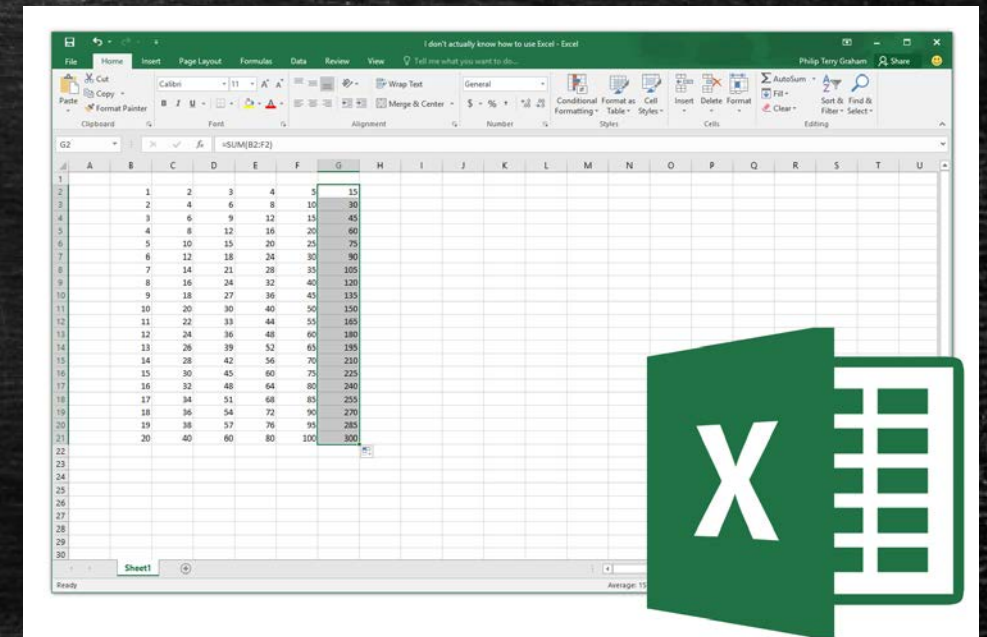
Overview

Introduction to Accessibility

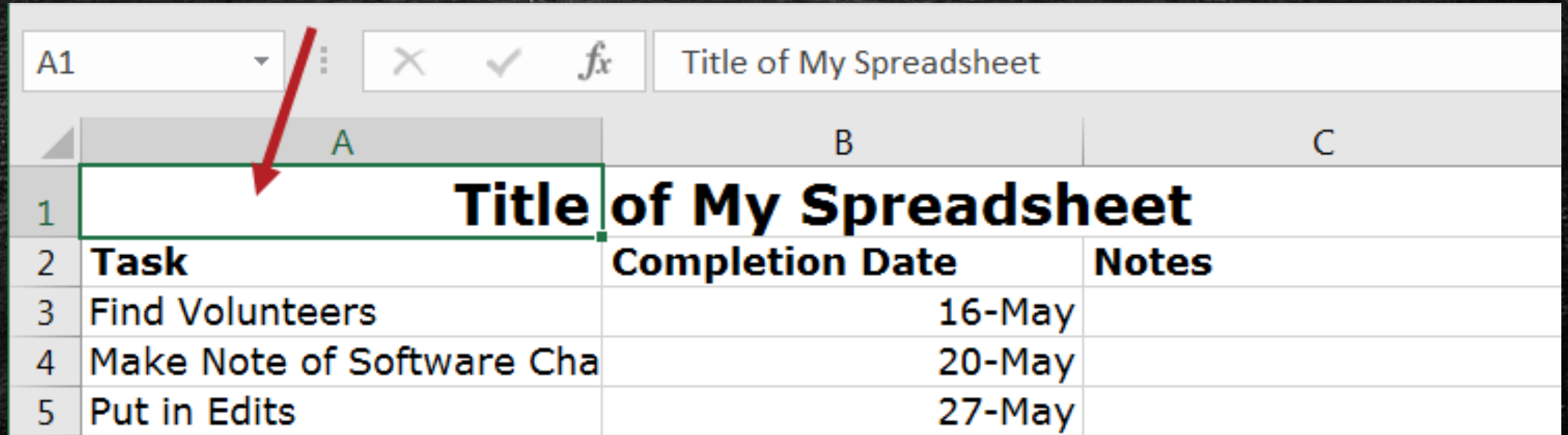


9 Basic Requirements for Excel

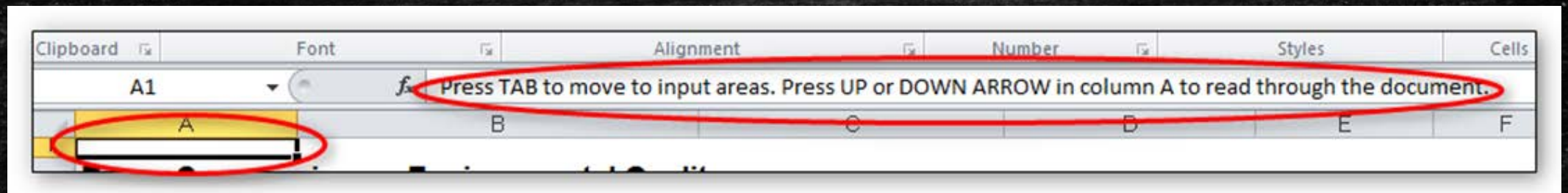
1. Start in A1
2. Stay in the Grid
3. No Blank Rows, Columns, Data Cells, and Hide Unused
4. No Merged or Split Cells in Data Portion
5. Worksheet Names and Table Titles; No Blank Worksheets
6. Row and Column Headers
7. Alternative Text
8. End of Worksheet, Workbook
9. Before Saving



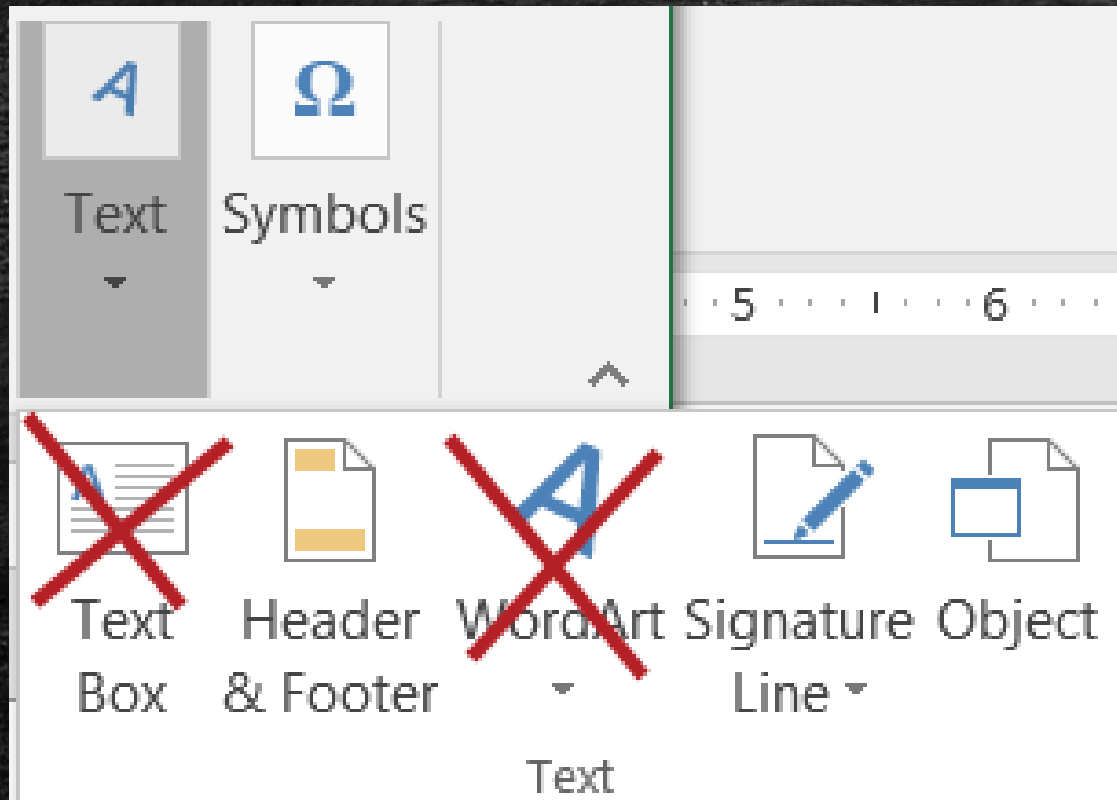
1. Start in A1



	A	B	C
1	Title of My Spreadsheet		
2	Task	Completion Date	Notes
3	Find Volunteers	16-May	
4	Make Note of Software Cha	20-May	
5	Put in Edits	27-May	

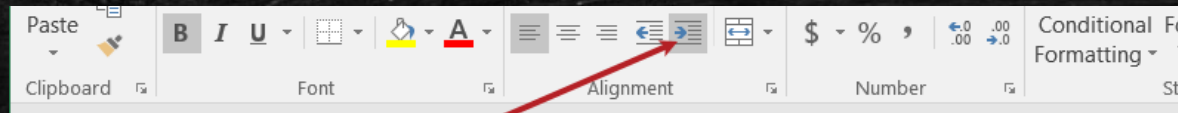


2. Stay in the Grid



3. No Blank Rows, Columns, Data Cells

A	B	C	D
Title of My Spreadsheet			
	Task	Completion Date	Notes
	Find Volunteers	16-May	
	Make Note of Software Cha	20-May	
	Put in Edits	27-May	
	Avoid		



A	B	C
Title of My Spreadsheet		
Task	Completion Date	Notes
Find Volunteers	16-May	
Make Note of Software Chan	20-May	
Put in Edits	27-May	

Indent text instead of inserting an empty column

	A	B	C
	Office 2013/2016 Accessibility Training Module		
1	Timeline - Summer 2016		
2	Task	Completion Date	Notes
3	Find Volunteers	16-May	No data
4	Look at old Scripts	20-May	No data
	Make note of		
5	Software changes	27-May	No data
6	Put in Edits	1-Jun	No data

Hide Unused Rows and Columns

A	B	C	D
Title of My Spreadsheet			
	Task	Completion Date	Notes
	Find Volunteers	16-May	
	Make Note of Software Cha	20-May	
	Put in Edits	27-May	
		Avoid	



1	A	B	C
2	Title of My Spreadsheet		
3	Task	Completion Date	Notes
4	Find Volunteers	16-May	
5	Make Note of Software Chan	20-May	
6	Put in Edits	27-May	

Indent text instead of inserting an empty column

	A	B	C
1	Office 2013/2016 Accessibility Training Module Timeline - Summer 2016		
2	Task	Completion Date	Notes
3	Find Volunteers	16-May	No data
4	Look at old Scripts	20-May	No data
5	Make note of Software changes	27-May	No data
6	Put in Edits	1-Jun	No data

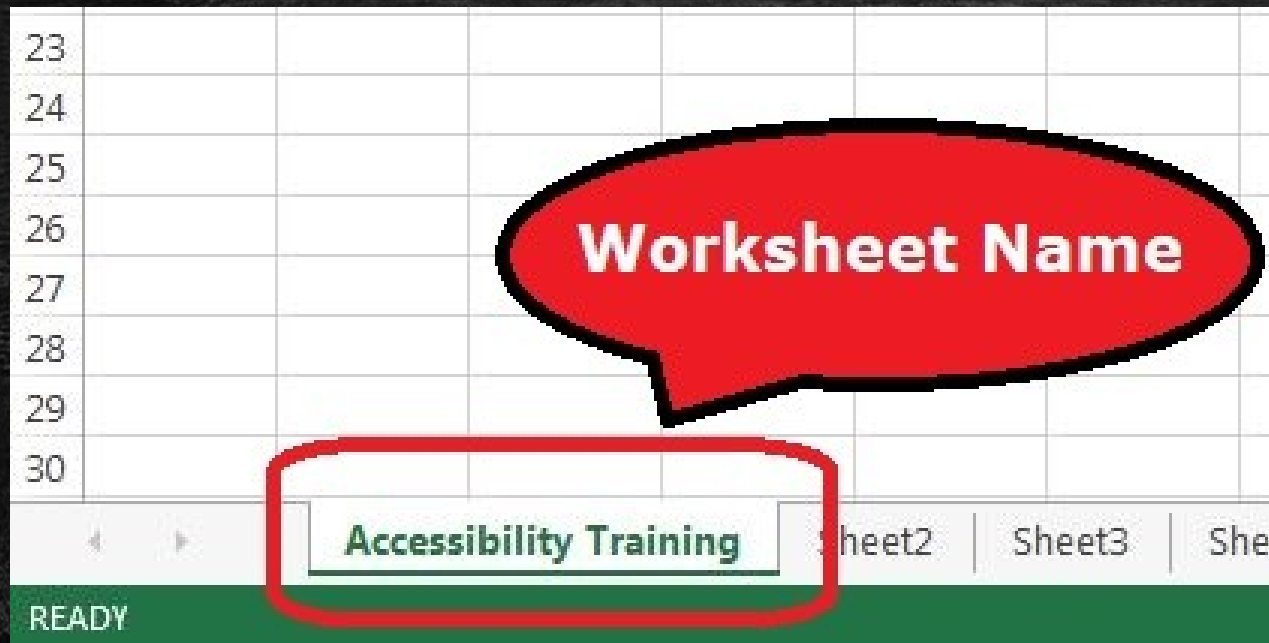
4. No Merged or Split Cells in Data Portion

	A	B	C
1	Title of My Spreadsheet		
2	Task	Completion Date	Notes
3	Find Volunteers	16-May	
4	Make Note of Software Changes	20-May	
5	Put in Edits	27-May	

	A	B	C	D	E
1	Test Results				
2		Test 1	Test 2		
3	Student 1	12	7		
4	Student 2	14	18		
5	Student 3	Missed both tests			
6	Student 4	11	5		

Merged

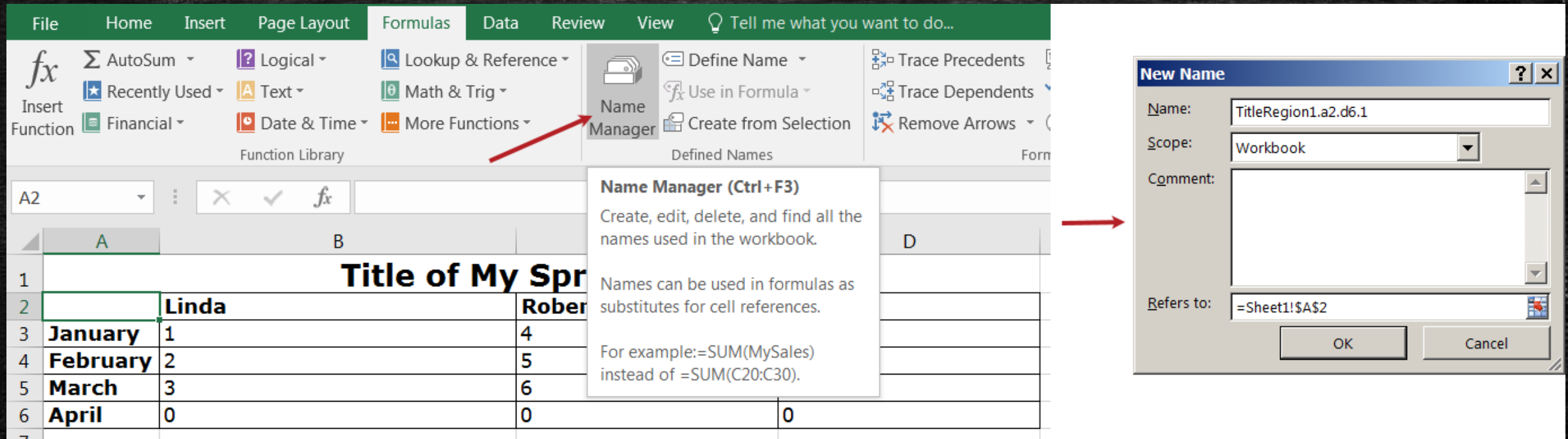
5. Worksheet Names, Table Titles, Blank Worksheets



Two screenshots of an Excel table. The top screenshot shows the table title "Summer Timeline" circled in red. The bottom screenshot shows the table title "Office 2013/2016 Accessibility Training Module Timeline - Summer 2016" circled in red. An arrow points from the top title to the bottom title.

Task	Completion Date	Notes
Find Volunteers	16-May	
Look at old Scripts	20-May	
Make note of		
Software changes	27-May	
Put in Edit	1-Jun	

6. Row and Column Headers



The screenshot illustrates the process of creating a named range in Excel. The **Formulas** ribbon is active, and the **Name Manager** button is highlighted with a red arrow. A tooltip for the Name Manager button is displayed, providing instructions on its use. To the right, the **New Name** dialog box is open, showing the configuration for a new named range.

Name Manager (Ctrl+F3)
Create, edit, delete, and find all the names used in the workbook.
Names can be used in formulas as substitutes for cell references.
For example: =SUM(MySales) instead of =SUM(C20:C30).

New Name

Name: TitleRegion1.a2.d6.1
Scope: Workbook
Comment:
Refers to: =Sheet1!\$A\$2

OK Cancel

	A	B	C	D
1	Title of My Spreadsheet			
2	Linda	Robert		
3	January	1	4	
4	February	2	5	
5	March	3	6	
6	April	0	0	0



Using Excel's Built-In Table Header – Slide 1 of 4

Only works with column headers

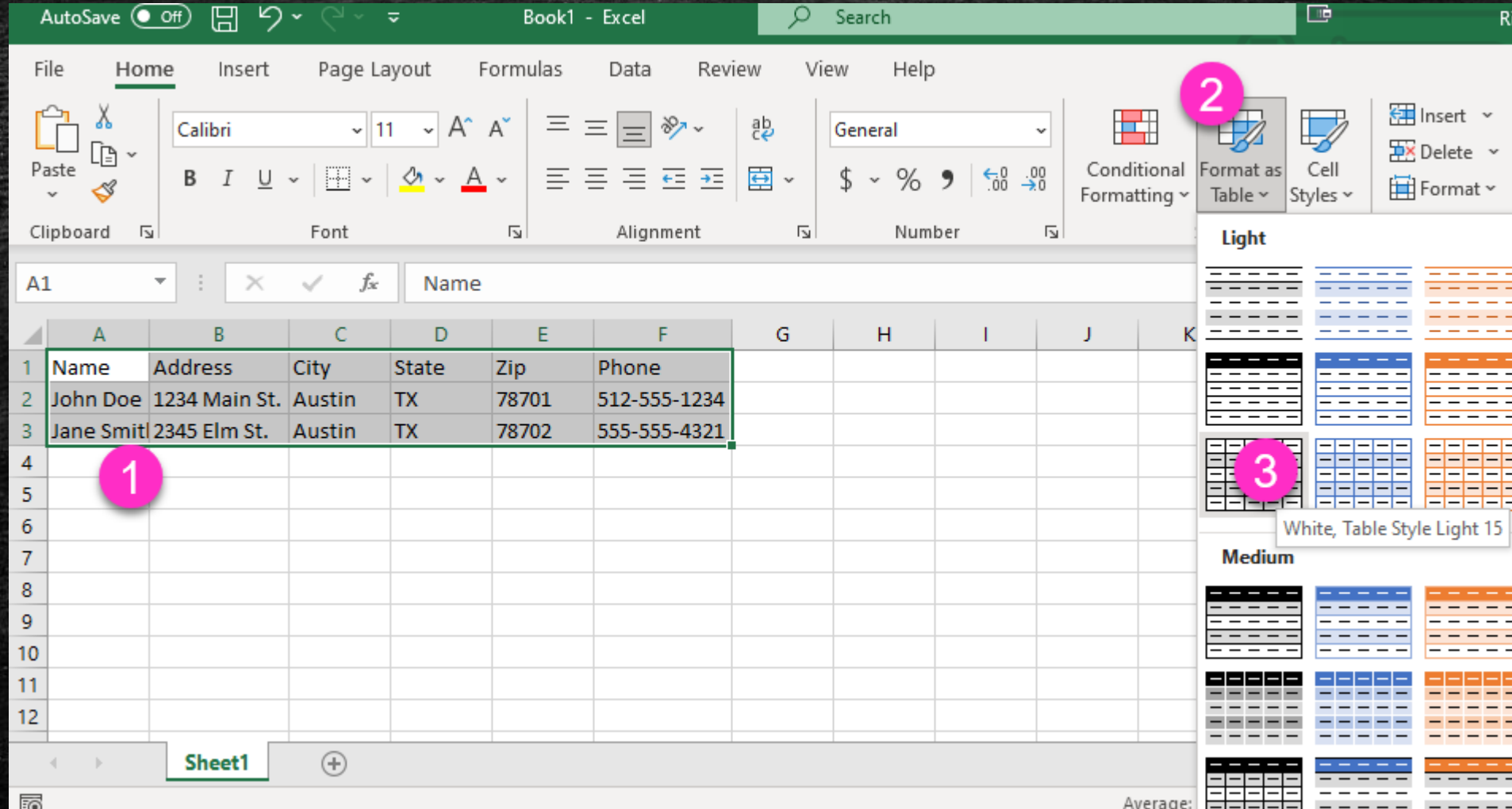
Works with JAWS 2019-2022 and Excel 2016-365

Doesn't work with NVDA or VoiceOver

[See video](#)

Using Excel's Built-In Table Header –

Slide 2 of 4



The screenshot shows the Microsoft Excel interface. The ribbon is set to 'Home'. The 'Format as Table' button is highlighted with a pink circle labeled '2'. The table in the worksheet is highlighted with a green border, and a pink circle labeled '1' is placed over the first row of data. The 'Format as Table' dropdown menu is open, showing various table styles. A pink circle labeled '3' is placed over a specific table style. The table in the worksheet has the following data:

Name	Address	City	State	Zip	Phone
John Doe	1234 Main St.	Austin	TX	78701	512-555-1234
Jane Smith	2345 Elm St.	Austin	TX	78702	555-555-4321

Using Excel's Built-In Table Header – Slide 3 of 4

The screenshot shows the Microsoft Excel interface with the 'Format As Table' dialog box open. The dialog box is titled 'Format As Table' and has a question mark icon and a close button. It asks 'Where is the data for your table?' and shows the range '\$A\$1:\$F\$3'. Below this, there is a checkbox labeled 'My table has headers' which is checked. At the bottom of the dialog box, there are 'OK' and 'Cancel' buttons. A pink circle with the number 4 is placed next to the 'Where is the data for your table?' section, and a pink circle with the number 5 is placed next to the 'OK' button.

Name	Address	City	State	Zip	Phone
John Doe	1234 Main St.	Austin	TX	78701	512-555-1234
Jane Smit	2345 Elm St.	Austin	TX	78702	555-555-4321

Using Excel's Built-In Table Header – Slide 4 of 4

- If you choose this method, give tables unique names

The screenshot shows the Microsoft Excel 365 interface. The ribbon is set to 'Table Design'. A pink circle with the number '1' highlights the 'Table Design' tab. Another pink circle with the number '2' highlights the 'Table Name' field, which contains 'MailingList'. The table below has columns: Name, Address, City, State, Zip, and Phone. The first two rows of data are highlighted.

AutoSave ☐ Off Test Excel 365 file.xlsx Search (Alt+Q)

File Home Insert Page Layout Formulas Data Review View Developer Help **Table Design**

Table Name: Resize Table Properties

Summarize with PivotTable Remove Duplicates Convert to Range Tools

Insert Slicer

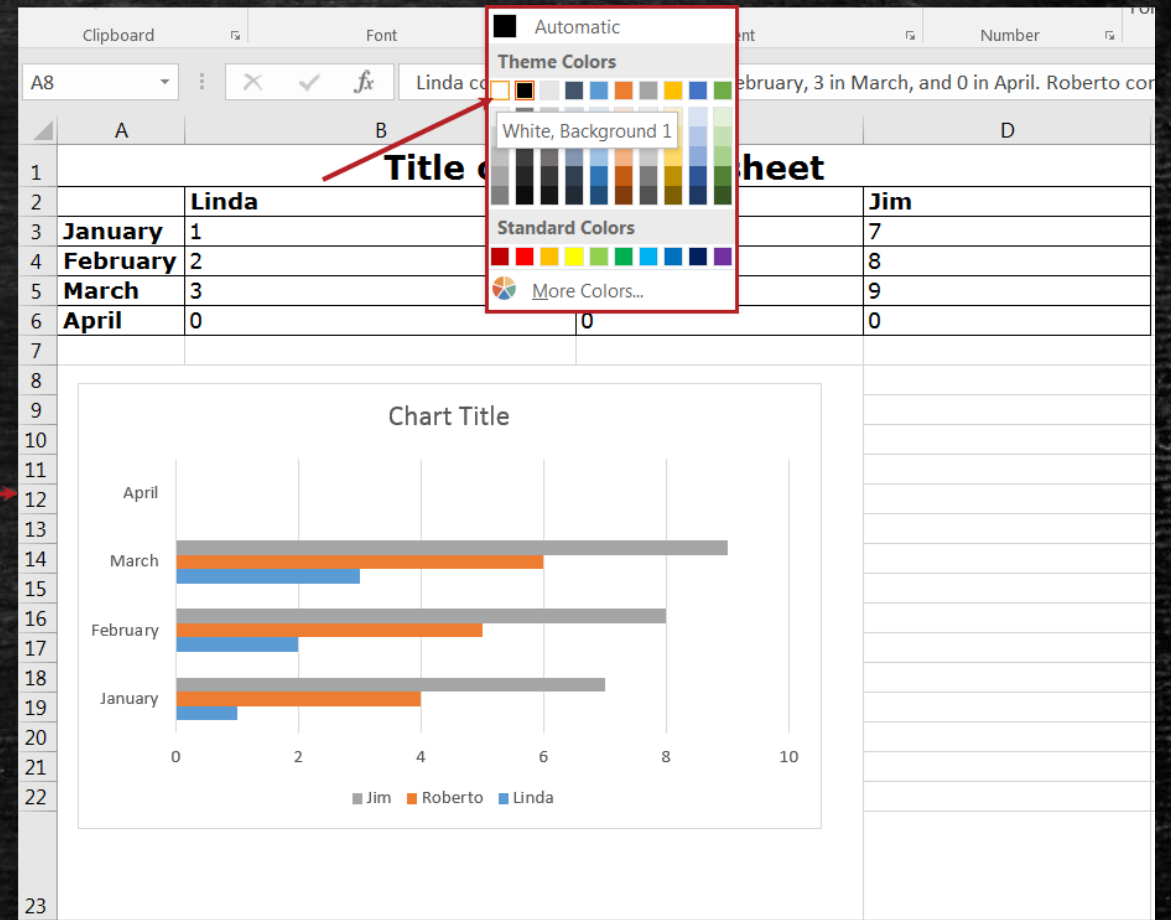
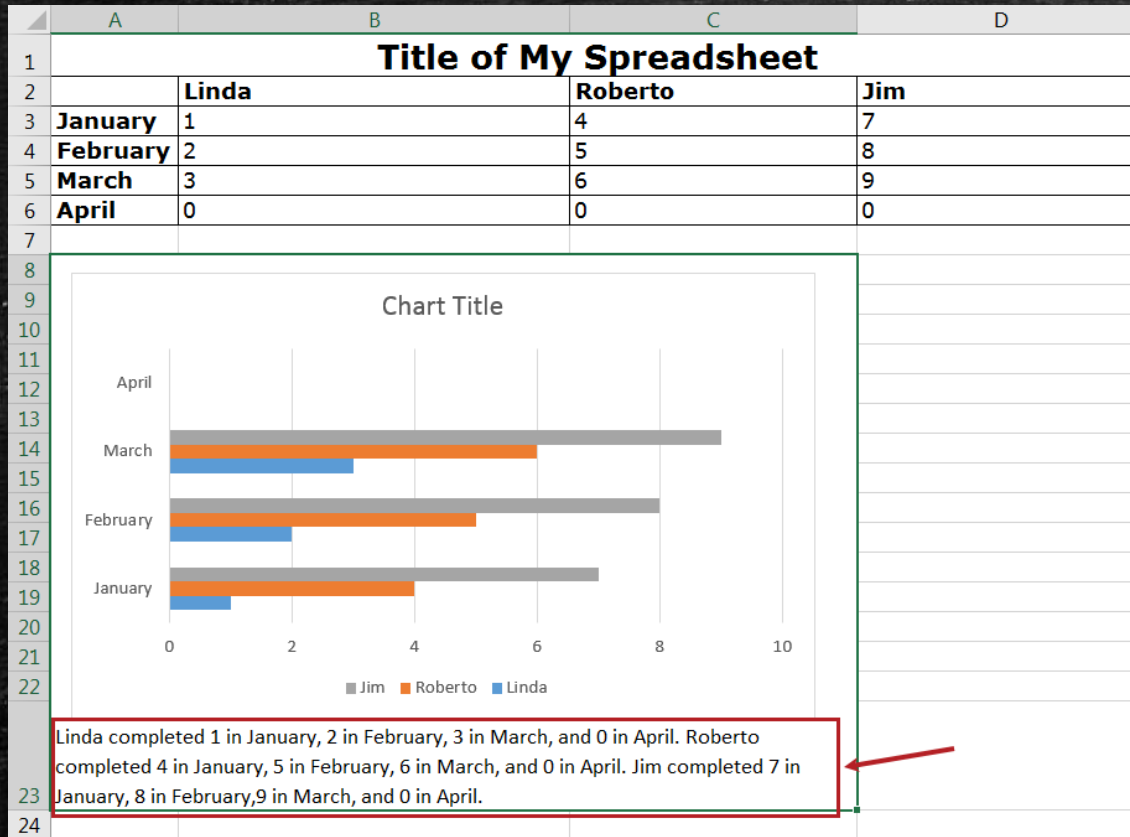
Export Refresh Properties Open in Browser Unlink External Table Data

☒ Header Row ☐ First Column ☐ Filter Button
☐ Total Row ☐ Last Column
☒ Banded Rows ☐ Banded Columns Table Style Options

A2 John Doe

	A	B	C	D	E	F
1	Name	Address	City	State	Zip	Phone
2	John Doe	1234 Main St.	Austin	TX	78701	512-555-1234
3	Jane Smith	2345 Elm St.	Austin	TX	78702	555-555-4321

7. Alternative Text



8. End of Worksheet, Workbook

	A	B	C
1	Title of My Spreadsheet		
2	Task	Completion Date	Notes
3	Find Volunteers	16-May	George will help
4	Make Note of Software Changes	20-May	Emailed IT department
5	Put in Edits	27-May	Schedule meeting
6	END OF WORKSHEET		
7			

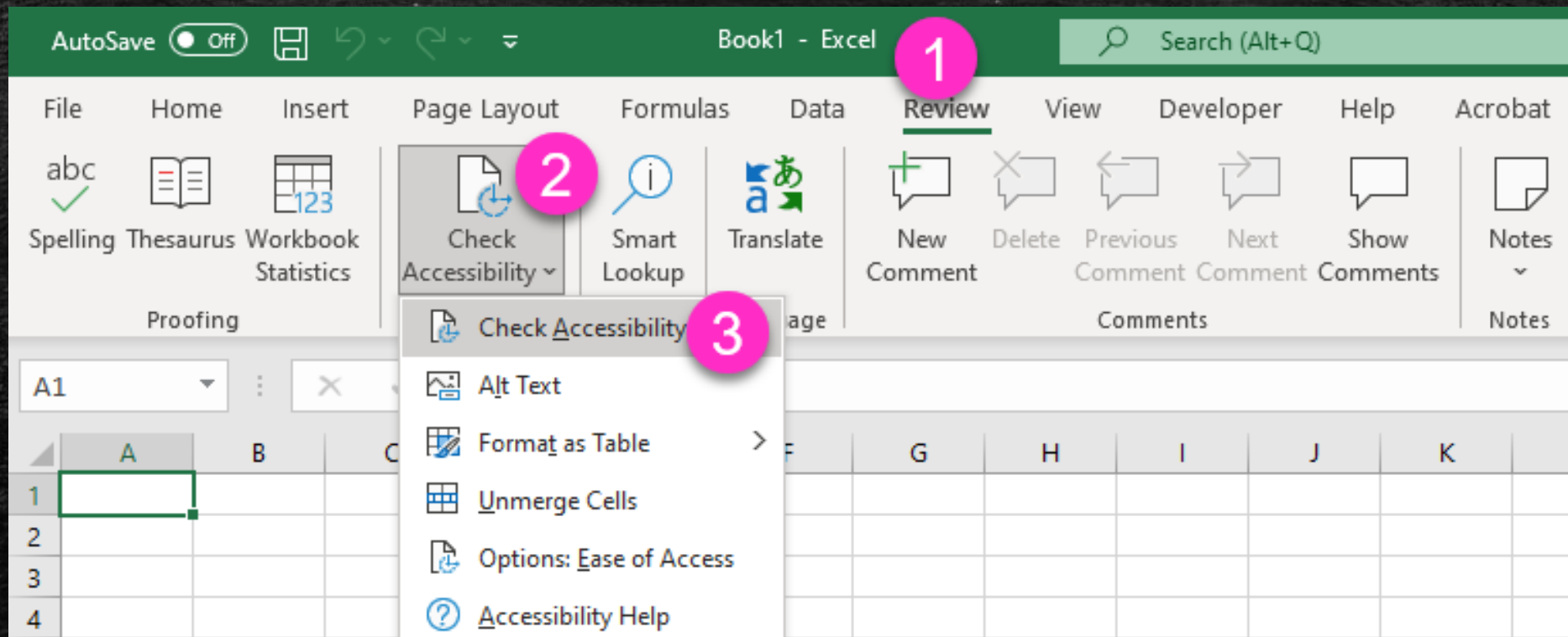


9. Before Saving

1. Select A1 cell on first sheet of workbook and save file unless you want users to start at a different part of the workbook when they open file
2. Check spelling
3. Set Print Area
4. Remove Comments
5. Run Accessibility Checker

More on Accessibility Checker, Slide 1

Accessed through the Review tab



More on Accessibility Checker – Slide 2

Alternate way to reach accessibility checker

The screenshot shows the Microsoft Excel ribbon with the 'Info' tab selected. The left sidebar contains navigation options: Home, New, Open, Info (highlighted with a pink circle and the number 1), Save, Save As, Print, Share, Export, Publish, Close, and Account. The main area displays the 'Info' tab for the file 'Simplify Tables Example'. It includes buttons for Upload, Share, Copy path, and Open file location. Below these are three main sections: 'Protect Workbook' (with a pink circle and the number 2), 'Inspect Workbook' (with a pink circle and the number 3), and 'Check Accessibility' (highlighted with a grey background). The 'Check Accessibility' section is expanded, showing a list of checks: 'Inspect Document' (Check the workbook for hidden properties or personal information), 'Check Accessibility' (Check the workbook for content that people with disabilities might find difficult to read), and 'Check Compatibility' (Check for features not supported by earlier versions of Excel).

Simplify Tables Example.x

Info

Simplify Tables Example

Documents » Accessibility » AccessU 2022

Upload Share Copy path Open file location

1 Info

Save

Save As

Print

Share

Export

Publish

Close

Account

2 Protect Workbook

Control what types of changes people can make to this workbook.

Protect Workbook

3 Inspect Workbook

Before publishing this file, be aware that it contains:

- Document properties, document server properties, content type information, printer path, author's name, related dates and absolute path

Check Accessibility

Check the workbook for content that people with disabilities might find difficult to read.

Check Compatibility

Check for features not supported by earlier versions of Excel.

More on Accessibility Checker, Slide 3

Will not flag lack of content in A1

Will not flag tables that are not defined, missing row/column headers

Will flag default sheet names, but doesn't flag undescriptive sheet names

Will flag missing alt text for images, but cannot evaluate quality of alt text

Will flag cell text with low contrast, but does not test charts, graphs and images

Excel Deficiencies – Slide 1

Formatting text using heading styles ignored by screen readers

The screenshot shows the Microsoft Excel interface. The ribbon is set to 'Home', and the 'Font' group is active. The task pane on the right is open, showing the 'Styles' section. A blue callout box with red text 'Screen readers ignore these' points to the 'Heading 1' style in the task pane. The spreadsheet shows a table with columns labeled 'Title' and 'Total'.

The Heading 1 styles					
20% - Accent1	20% - Accent2	20% - Accent3	20% - Accent4	20% - Accent5	20% - Accent6
40% - Accent1	40% - Accent2	40% - Accent3	40% - Accent4	40% - Accent5	40% - Accent6
60% - Accent1	60% - Accent2	60% - Accent3	60% - Accent4	60% - Accent5	60% - Accent6
Accent1	Accent2	Accent3	Accent4	Accent5	Accent6

Excel Deficiencies – Slide 2

Converting Excel to accessible PDF is near impossible

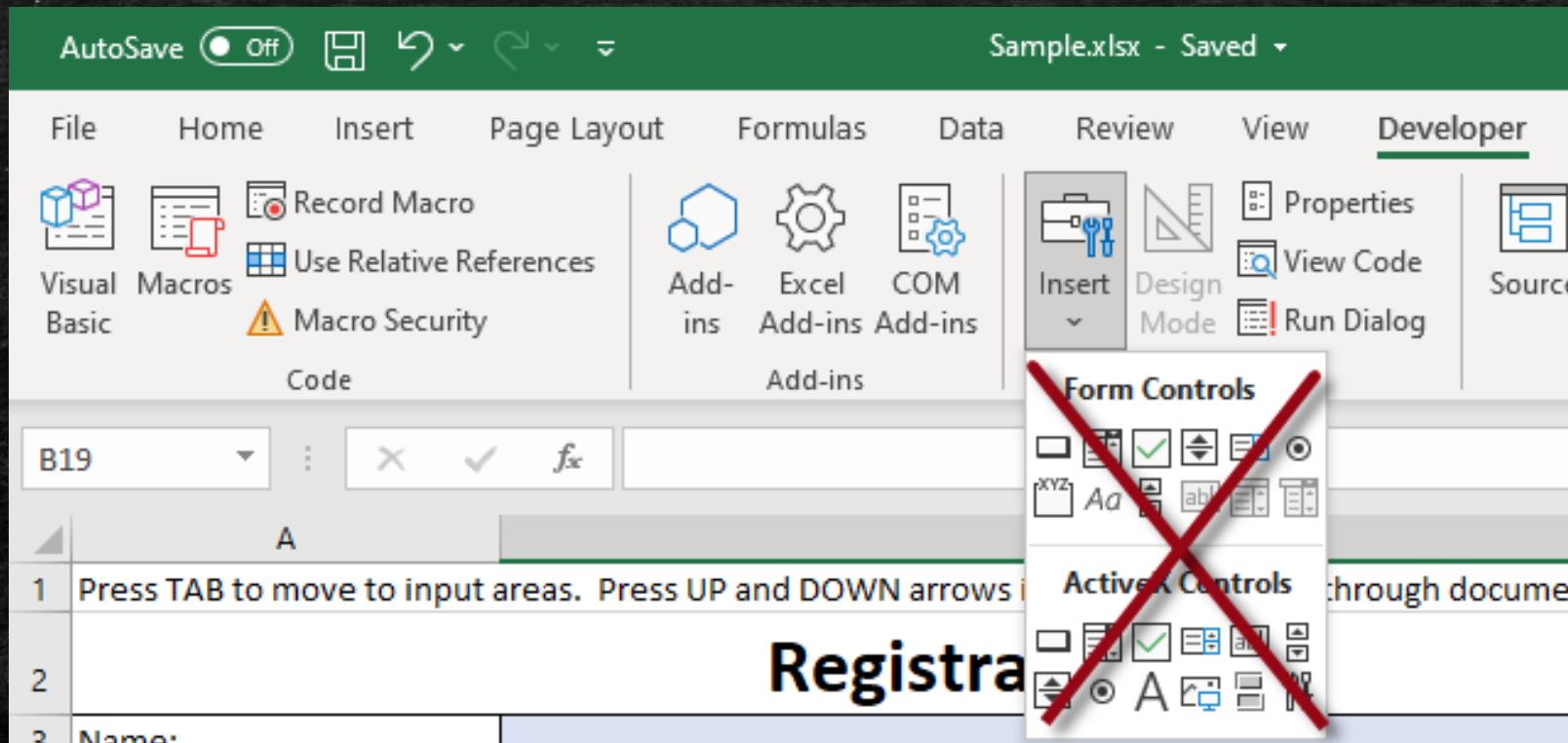
	A	B	C	D
1	Department of Human Services			
2	FY 2020 Monthly Financial Report: Capital Budget			
3	Data through March 2020			
4	Capital Projects in Capital Rider	Appropriated	Op. Bgt.	Projected
5	53001 BUILDING MAINTENANCE	\$ 345,000	\$ 345,000	\$ 345,000
6	53002 SEAT MANAGEMENT SERVICES	\$ 2,000,000	\$ 2,000,000	\$ 2,000,000
7	53003 COMPLIANCE WITH FERDERAL HIPAA	\$ 35,690	\$ 35,690	\$ 35,690
8	53004 FOOD SERVICE FACILITIES	\$ 200,000	\$ 200,000	\$ 200,000
9	53011 REHABWORKS	\$ -	\$ 542,317	\$ 542,317
10	53150 DATA CENTER CONSOLIDATION	\$ 4,444,272	\$ 2,949,136	\$ 2,949,136
11	GRAND TOTAL, DHS	\$ 7,024,962	\$ 6,072,143	\$ 6,072,143

Tags

- <Document>
- <Sect>
- <Table>
- <TR>
- <TR>
- <TR>
- <Table>
- <UnknownNodeType> ?
- <TR>
- <UnknownNodeType> ?
- <TR>
- <TR>
- <TR>
- <TR>

Excel Forms, Step 1

Form fields cannot be used because they lay on top of the surface of the spreadsheet and can't be reached by screen reader



Excel Forms, Basic Design

Designing Forms

	A	B
1	Press TAB to move to input areas. Press UP and DOWN arrows in column A to read through document.	
2	Registration	
3	First Name:	
4	Last Name:	
5	Street Address 1:	
6	Street Address 2:	
7	City	
8	State	
9	ZIP Code	
10	Email:	

Excel Forms, Design Enhancements

Shade Input Cells and Add Border Lines for Visual Effect

	A	B
1	Press TAB to move to input areas. Press UP and DOWN arrows in column A to read through document.	
2	Registration	
3	First Name:	
4	Last Name:	
5	Street Address 1:	
6	Street Address 2:	
7	City	
8	State	
9	ZIP Code	
10	Email:	

Excel Forms, Data Validation

Sample.xlsx - Saved

File Home Insert Page Layout Formulas **Data** Review View Help Acrobat

Get & Transform Data: Get Data (From Text/CSV, From Web, From Table/Range), Recent Sources, Existing Connections, Refresh All

Queries & Connections: Queries & Connections, Properties, Edit Links

Data Types: Stocks, Currencies

Sort & Filter: Sort, Filter, Clear, Reapply, Advanced

Text Columns: Flash Fill, Remove Duplicates, Data Validation (circled with pink 2), Circle Invalid Data (circled with pink 3), Clear Validation Circles

B3: Registration (circled with pink 1)

1	Press TAB to move to input areas. Press UP and DOWN arrows in column A to read through document.
2	
3	Name:
4	Street Address 1:
5	Street Address 2:
6	City
7	State
8	ZIP Code
9	Email:

Sample.xlsx - Saved

File Home Insert Page Layout Formulas **Data** Review View Help Acrobat

Get & Transform Data: Get Data (From Text/CSV, From Web, From Table/Range), Recent Sources, Existing Connections, Refresh All

Queries & Connections: Queries & Connections, Properties, Edit Links

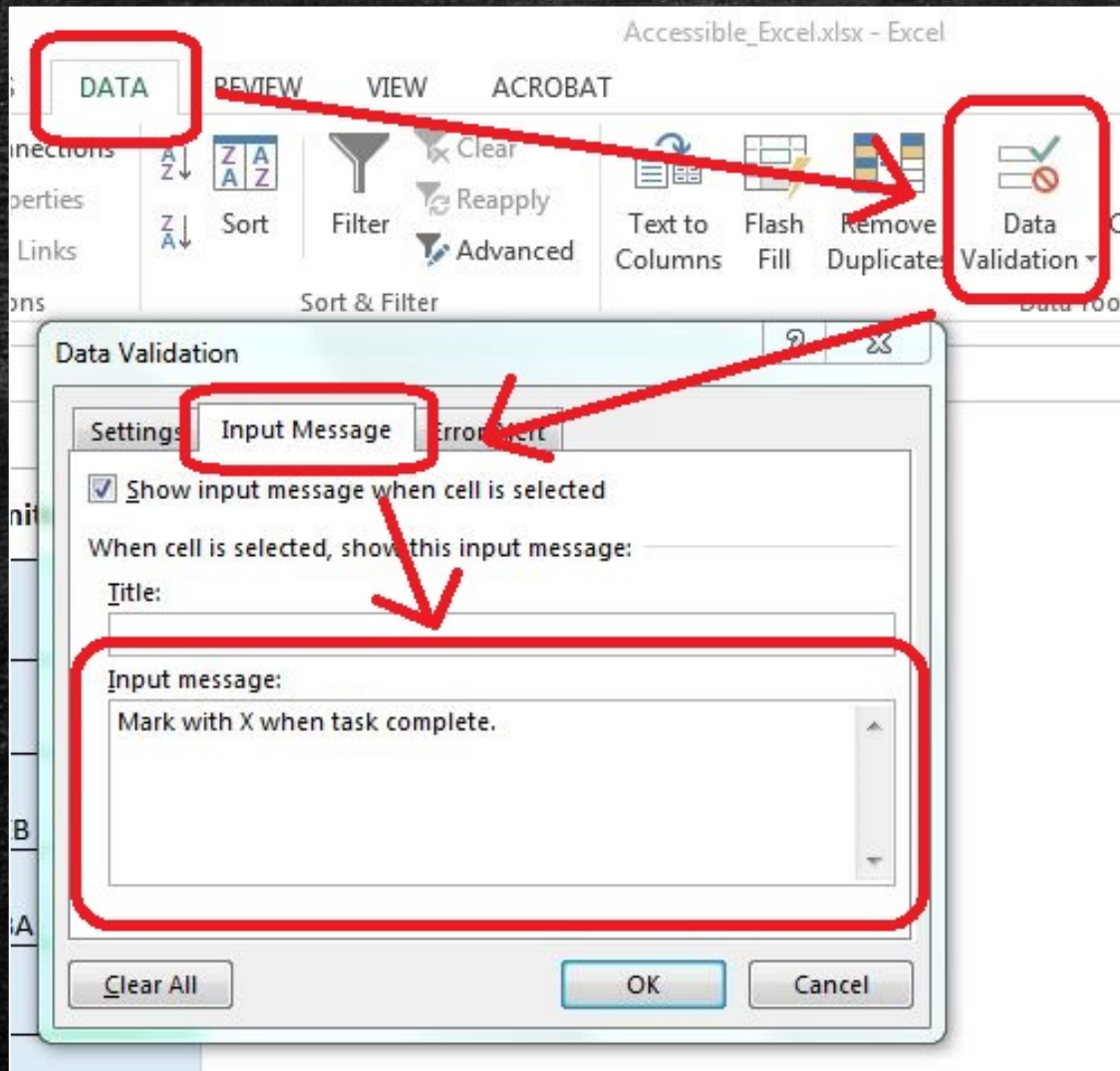
Data Types: Stocks, Currencies

Data Validation dialog box:

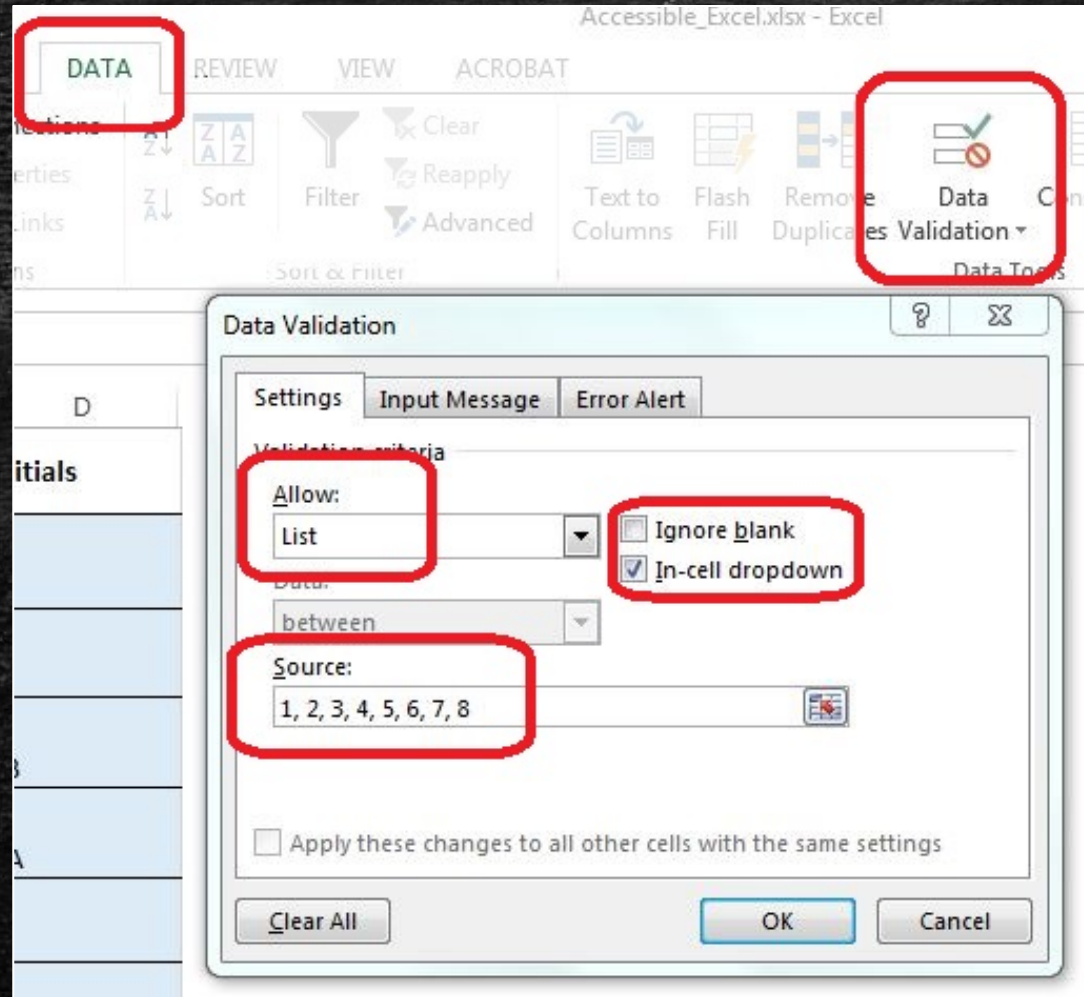
- Settings tab
- Input Message tab (selected)
- Error Alert tab
- 4: Show input message when cell is selected
- 5: When cell is selected, show this input message:
- 6: Input message: Enter in format last name, first name
- 7: OK button

1	Press TAB to move to input areas. Press UP and DOWN arrows in column A to read through document.
2	
3	Name:
4	Street Address 1:
5	Street Address 2:
6	City
7	State
8	ZIP Code
9	Email:

Excel Forms, Data Validation Example 2



Excel Forms, Creating Drop Down Lists, Part 1

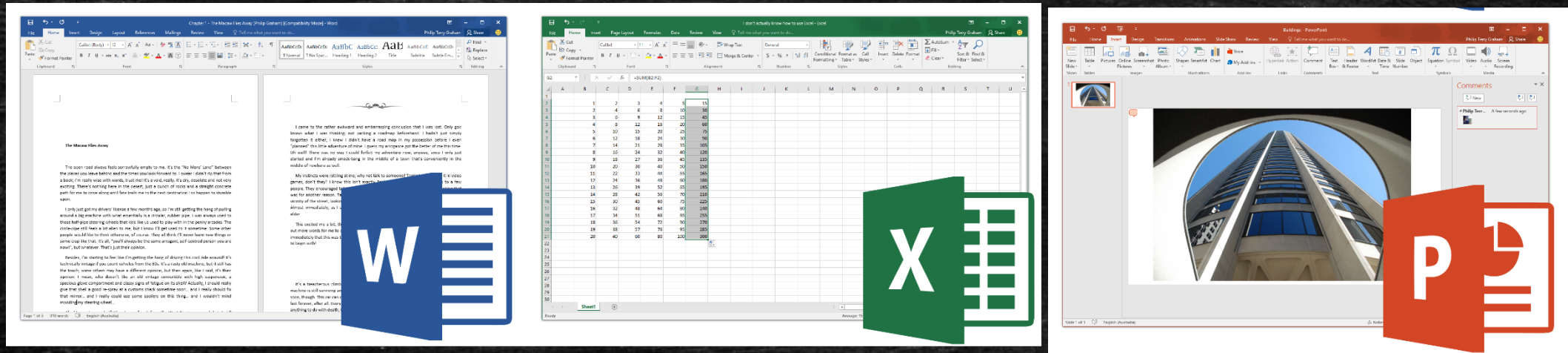




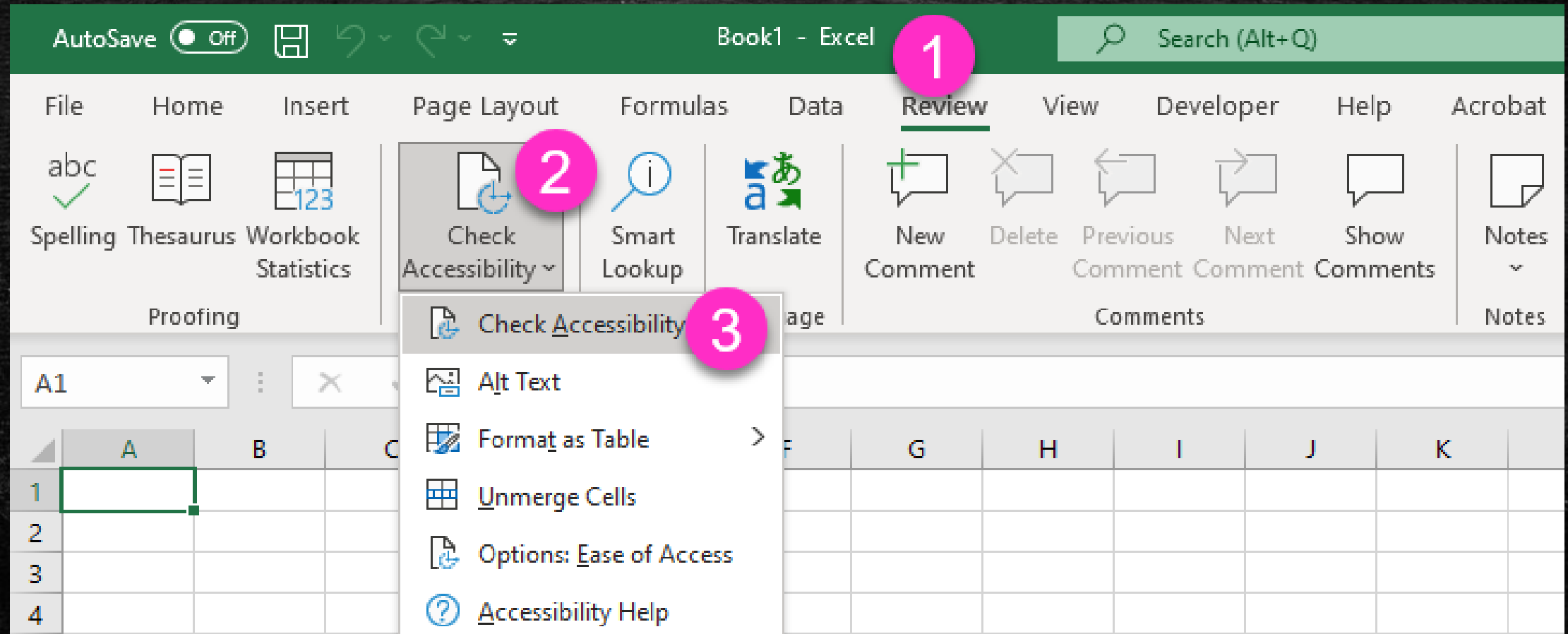
Put in number
of tasks
complete.

21

Tips for ALL MS Office



Using MS Office 365 Accessibility Checker



Using MS Office 2016 Accessibility Checker

The screenshot displays the MS Office 2016 interface with the 'Exercise.xlsx - Excel' window. The left sidebar contains navigation options: Info, New, Open, Save, Save As, History, Print, Share, Export, Publish, Close, Account, and Options. The main area shows the 'Info' tab for the 'Exercise' workbook, located at 'Desktop » IL-ates » Excel'. It includes sections for 'Protect Workbook', 'Inspect Workbook', 'Properties', 'Related Dates', and 'Related People'. The 'Inspect Workbook' section is expanded, showing options to 'Inspect Document', 'Check Accessibility', and 'Check Compatibility'. The 'Check Accessibility' option is highlighted, indicating it has been selected. The right pane, titled 'Accessibility Checker', displays the 'Inspection Results' for the selected option. It lists two errors: 'Missing Alt Text' for 'Picture 1' and 'No Header Row Specified' for a 'Table'. It also lists two warnings: 'Merged or Split Cells' for a 'Table Cell' and 'Repeated Blank Characters' for '9 Characters'.

Accessibility Checker ▾ ✕

Inspection Results

Errors

- ▲ Missing Alt Text
Picture 1
- ▲ No Header Row Specified
Table

Warnings

- ▲ Merged or Split Cells
Table Cell
- ▲ Repeated Blank Characters
9 Characters

Document Properties

The image illustrates the process of accessing document properties in Microsoft Excel. It shows the 'Info' ribbon, the 'Properties' pane, and the 'Accessible_Excel.xlsx Properties' dialog box.

Excel Interface:

- Info Ribbon:** The 'Info' tab is selected in the ribbon. It contains sections for 'Protect Workbook' and 'Inspect Workbook'.
- Properties Pane:** The 'Properties' pane is open on the right side of the ribbon. It shows the 'Advanced Properties' link, which is highlighted with a red box. A red arrow points from the 'Info' tab to this link.
- Document Properties Dialog Box:** The 'Accessible_Excel.xlsx Properties' dialog box is open, showing the 'General' tab. It contains fields for Title, Subject, Author, Manager, Company, Category, Keywords, Comments, Hyperlink base, and Template. The 'Save Thumbnails for All Excel Documents' checkbox is also visible.

Properties Pane Details:

Properties

- Show Document Panel
Edit properties in the Document Panel.
- Advanced Properties**
See more document properties

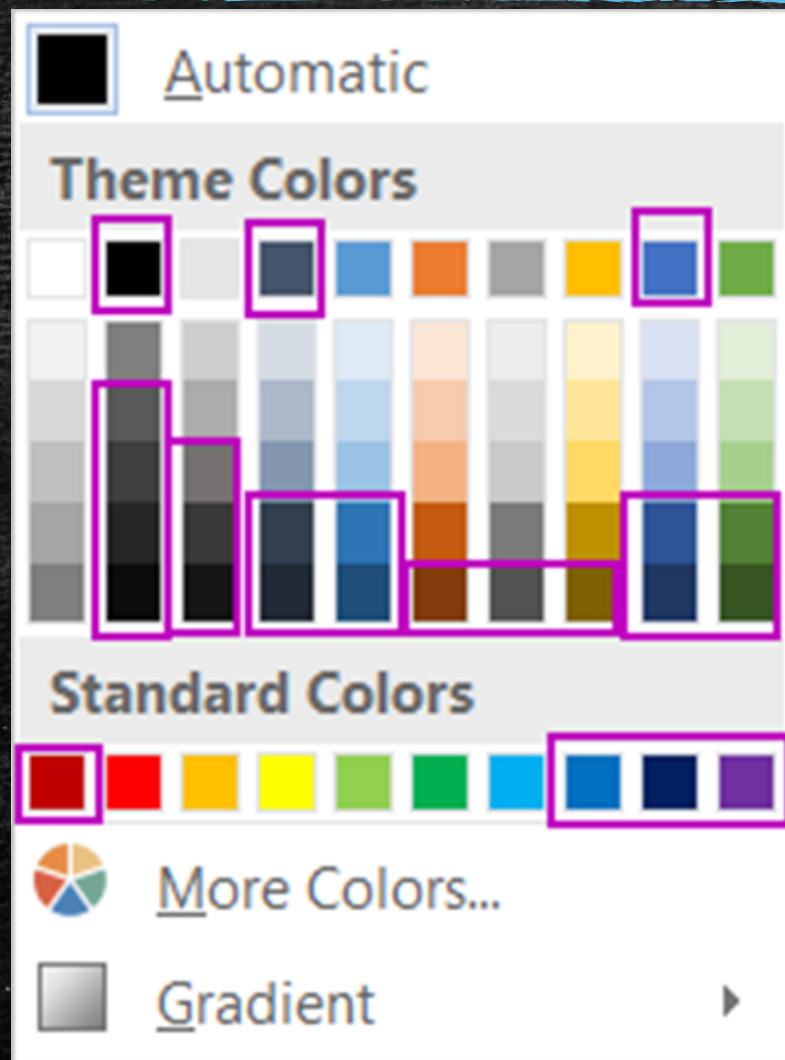
Document Properties:

Property	Value
Size	11.6KB
Title	Add a title
Tags	Add a tag
Categories	Add a category

Related Dates:

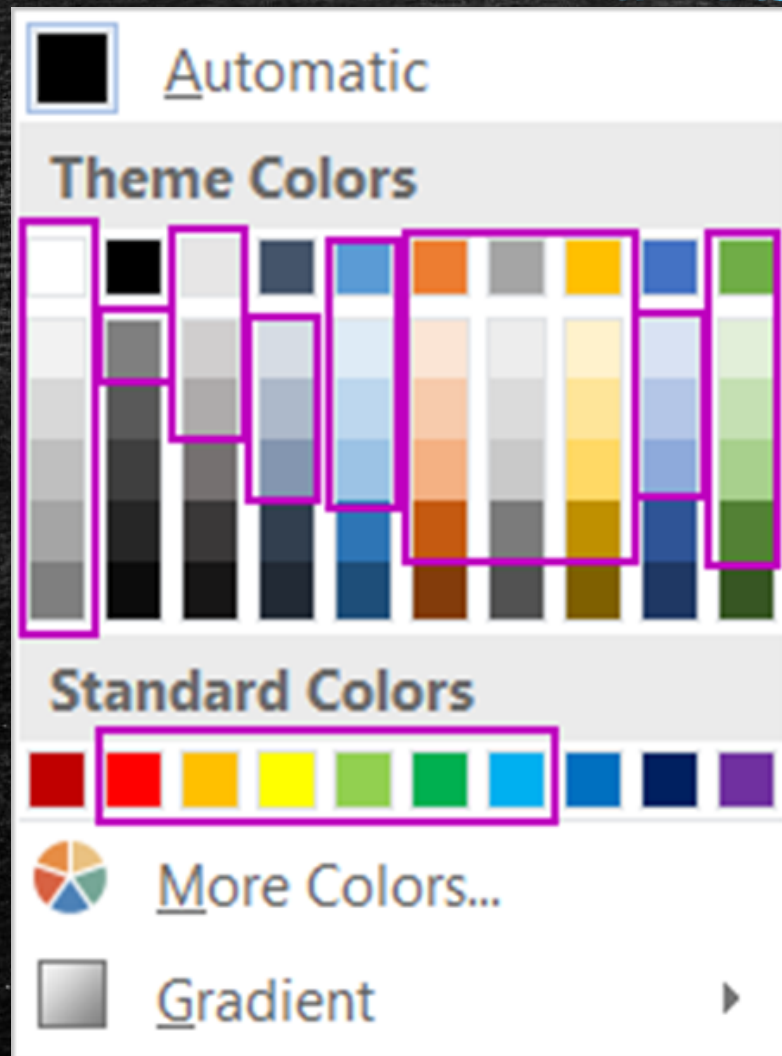
Property	Value
Last Modified	Today, 4:27 PM
Created	Today, 3:34 PM
Last Printed	

Colors that work on white



The colors that are boxed work on a white background

Colors that work on black



The colors that are boxed work on a black background

Colour Contrast Analyser

<https://www.tpgi.com/color-contrast-checker/>

The screenshot shows the Colour Contrast Analyser (CCA) web application. The interface is divided into several sections:

- Foreground colour:** Labeled "(black)", it features a "HEX" dropdown menu and a text input field containing "#000000". A yellow arrow points to the input field. Below the input field are icons for swap, copy, paste, and help.
- Background colour:** Labeled "(white)", it features a "HEX" dropdown menu and a text input field containing "#FFFFFF". A red arrow points to the copy icon below the input field. Below the input field are icons for swap, paste, and help.
- Sample preview:** A section with a dropdown arrow and the text "example text showing contrast" inside a dashed box. To the right is a bar chart icon.
- WCAG 2.1 results:** A section showing the "Contrast ratio" as "21:1". It lists three criteria:
 - 1.4.3 Contrast (Minimum) (AA):** Shows two green checkmarks and the text "Pass (regular text)" and "Pass (large text)". A red arrow points to the first checkmark.
 - 1.4.6 Contrast (Enhanced) (AAA):** Shows two green checkmarks and the text "Pass (regular text)" and "Pass (large text)".
 - 1.4.11 Non-text Contrast (AA):** Shows one green checkmark and the text "Pass (UI components and graphical objects)".

Questions?

???

Thank you